

# ECEN-607 Final Project

## A 10-bit 50MS/s SAR-ADC in 90nm Technology

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**Abstract** – A 10-bit 50MS/s fully differential SAR-ADC in 90nm technology is presented. The design employs the gate-bootstrapped sampling switch, StrongArm Comparator, and split-CDAC array which was switched in monotonic way to improve energy efficiency. An asynchronous control logic is implemented on transistor level to enhance the conversion speed. At 1.2-V supply and 50MS/s sampling rate, the ADC achieves SNR of 59.98 dB which corresponds to ENOB of 9.55, with total power consumption of 1.31mW, achieving Schreier FOM of 163dB.

The rest of the report is organized as follows: Section-I describes System Architecture. Section-II describes block-level design of the comparator, switches, flipflops & other logic circuitry. Section-III describes top-level schematic and testbench. Section-IV contains the simulation results. Conclusions are given in Section V.

### I. SYSTEM ARCHITECTURE

SAR-ADC involves four major sections: sampling switch, comparator, CDAC array, and control logic. In this section, the details of timing and the events occurring during one complete clock period (acquisition + conversion phases) are described.

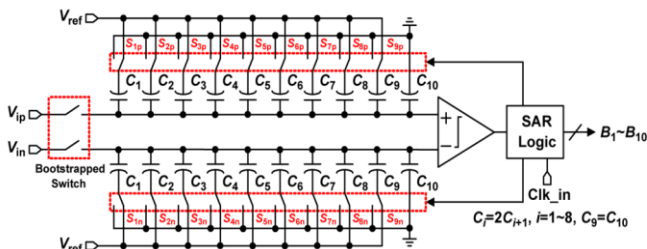


Fig1. Monotonic Switching Based 10-bit SAR-ADC

#### A. Monotonic Switching and DAC structure:

In a differential SAR-ADC we must digitize the  $V_{ip}$ - $V_{im}$  difference.

$$V_{ip} = V_{id}/2 - V_{rd}/2 + V_{CM}$$

$$V_{im} = -V_{id}/2 + V_{rd}/2 + V_{CM}$$

Where  $V_{rd} = K \cdot V_{ref}/(2^{(N-1)})$ ; N represents the no. of bits

$\Rightarrow V_{ip} = V_{id}/2 - K \cdot V_{ref}/(2^N) + V_{CM}$  which implies there will  $2^N$  cap requirement on both +ve and -ve side. So total Cap of  $2^{(N+1)}$

To reduce the total number of capacitances, comparison was not done in truly differential fashion.

$K > 0$ :

$$V_{ip} = V_{id}/2 - K \cdot V_{ref}/(2^{(N-1)}) + V_{CM}$$

$$V_{im} = -V_{id}/2 + V_{CM}$$

$K < 0$ :

$$V_{ip} = V_{id}/2 + V_{CM}$$

$$V_{im} = -V_{id}/2 - K \cdot V_{ref}/(2^{(N-1)}) + V_{CM}$$

This approach requires total of  $2^N$  cap ( $2^{(N-1)}$  on each +ve and -ve side).

In the monotonic switching approach, once a capacitance is switched to GND (from VREF as initially all the caps are connected to VREF), its not switched back to VREF again based on the comparison results, some other cap is switched to meet reference voltage requirements. This ensures SAR-ADC is running in an energy efficient manner.

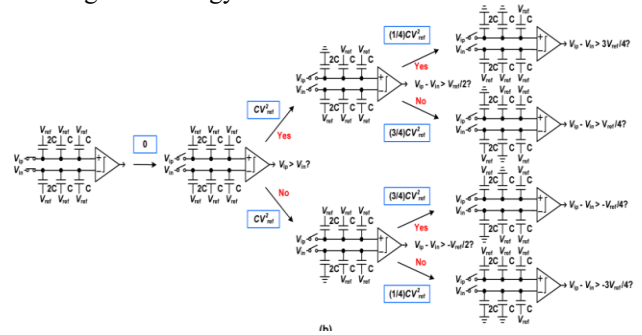


Fig2. Monotonic Switching Scheme (3-bit SAR-ADC example)

#### B. Asynchronous Logic:

Once the comparison is done comparators generate an internal signal (NAND of  $Q_p$  &  $Q_m$ , where  $Q_p$  &  $Q_m$  represents the output of the comparator). There is ring counter that is being operated using this signal, based on the ring counter output appropriate D-Flipflop is chosen for storing the comparison data. For example: The first comparison corresponds to MSB, D-Flipflop with respect to the MSB bit is clocked using the ring

counter output. At the start of every clock cycle counter is cleared as can be seen below.

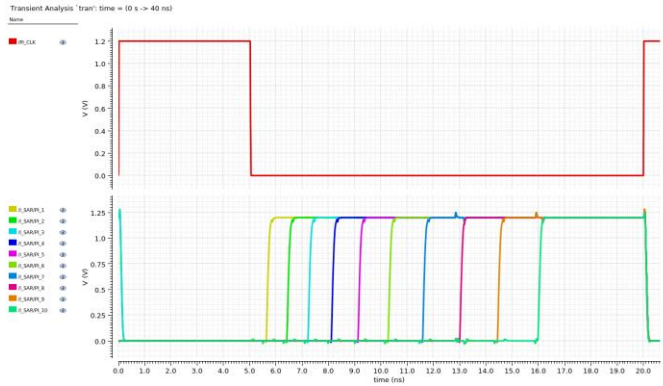


Fig3. Asynchronous logic signals

### C. Choice of unit Capacitor & Split C-DAC concept:

$C_{Total} = 2^N * C_u$  where  $C_u$  represents the unit capacitor. Capacitor seen by each switch during the acquisition phase is  $2^{(N-1)} * C_u$ .

Let's say  $C_s = 2^{(N-1)} * C_u$

In every process there is minimum capacitor that can be realized, which generally ensures that  $KT/C_s$  sampled thermal noise is way less than quantization noise and  $SNDR \sim SQNR$  which is given by  $6.02*N+1.76$  dB. But the effective cap that we get using this minimum capacitor generally is of very higher order as compared to the capacitance required for  $KT/C_s$  noise to be equal to quantization noise which means capacitive DAC network is overdesigned. To fix this problem split CDAC scheme is being used which reduces the effective capacitance seen by the sampling switch using the same unit-capitance. It was also ensured that unit capacitor is atleast 10 times larger than input capacitance of the strongArm comparator.

Split-CDAC scheme is primarily used to reduce the amount of capacitance experienced by the sampling switch, which helps to reduce the acquisition time and hence increase the sampling rate.

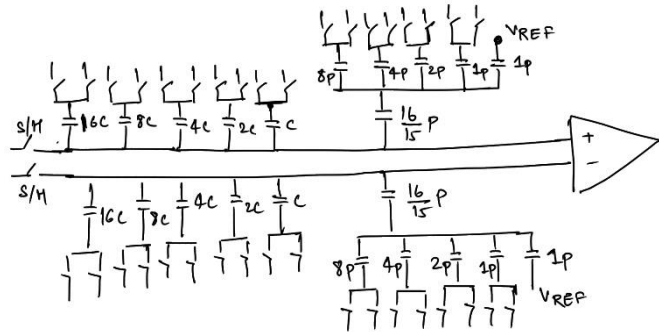


Fig4. Split C-DAC Scheme

This scheme reduces the effective capacitance seen by sampling switch to 32pF, considering the unit cap as 1pF. This greatly reduces the acquisition time and hence increases the sampling rate.

The Key issue with split C-DAC scheme is realizing the bridge capacitor which is not an integer multiple of unit capacitor.

### D. Complete Timing over one Clock-Cycle:

The S/H circuit was slightly oversized, it gave  $R_{on} \sim 40\Omega$ . Total Cap which each S/H observe is around 32pF, which leads to time constant  $\sim 0.128ns$ .

Acquisition Time  $> 7*\tau \sim 0.9ns$

Acquisition Time was kept as  $0.25*(T_{s\_CLK})$  where  $T_{s\_CLK}$  represents the clock period which is 20ns for 50MHz clock.

Acquisition Time =  $0.25*20ns = 5ns$

Conversion Time =  $0.75*20ns = 15ns$

All the 10 comparisons happen during the conversion period of 15ns. (Strongarm comparator was designed by keeping this thing in mind).

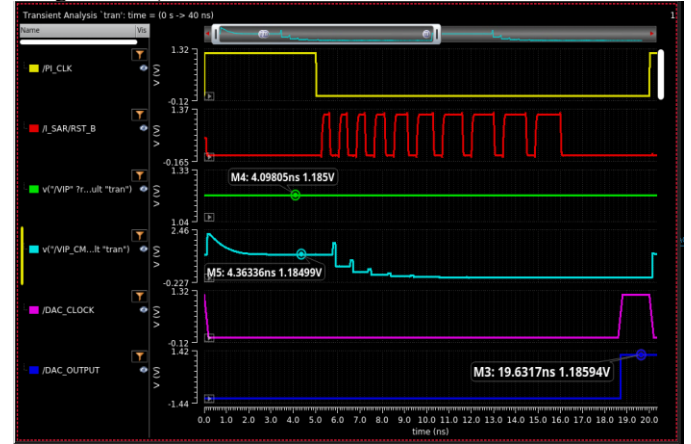


Fig5. SAR-ADC one Cycle for input differential voltage :1.185

Step1: Acquisition Phase, Here the S/H circuit samples the input on the capacitors (CDAC array). All the capacitors in the CDAC are connected to VREF in this phase.

Step2: Conversion Cycle, Comparator is clocked by inverted sampling clock signal which means comparison start as soon as the acquisition phase ends.

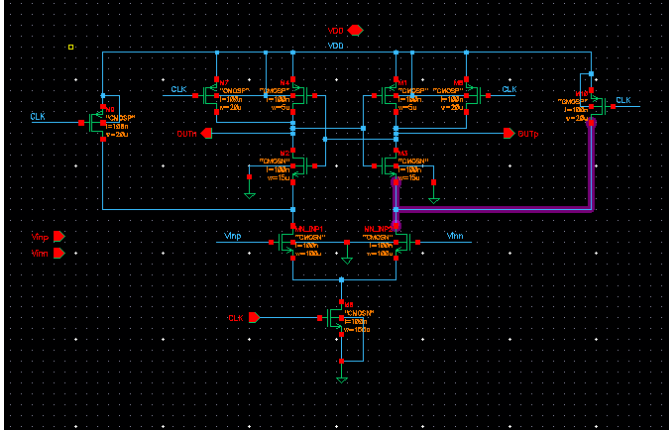
Step3: After the First comparison, internal signal is generated which indicates that the comparison is finished, now there is some delay introduced using the chain of inverters before the comparator resets and outputs are in its precharged state (supply in this case). During this time 1) Ring counters generate appropriate signals to tell which D-FlipFlop should store the current comparison results, 2) CDAC is settled to reference based on the comparison results.

Step3 Happens N-Times (10 in this case)

Step4: Once the comparison is done, all the bits  $D<9:0>$  are stored in the FlipFlop then ideal DAC converts these bits into analog value which stays in the hold state till the next cycle. At the start of every new cycle all the flipflops (ring counters and the storage) are cleared. (RESET of the D FlipFlop is driven by the clock signal).

## II. BLOCK LEVEL DESIGN

### A. StrongArm Comparator

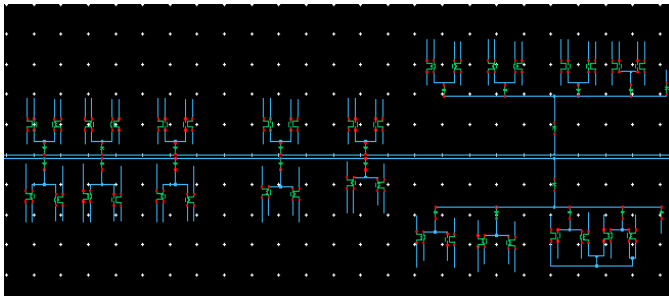


I/P Pair W/L = 100u/0.1u

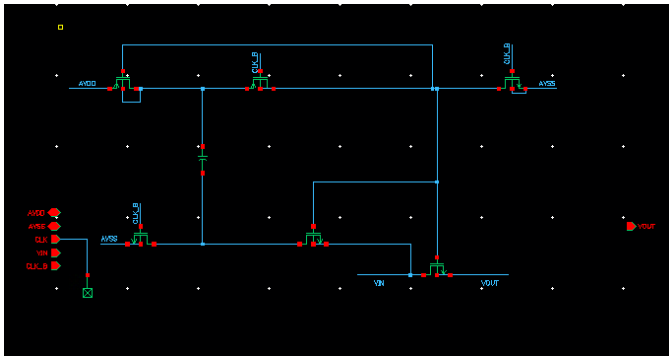
Bottom Current Source = 100u/0.1u

Higher the input pair size, lesser is the input referred offset but higher parasitic capacitance. It was ensured that this capacitance is less than  $0.1C_u$ .

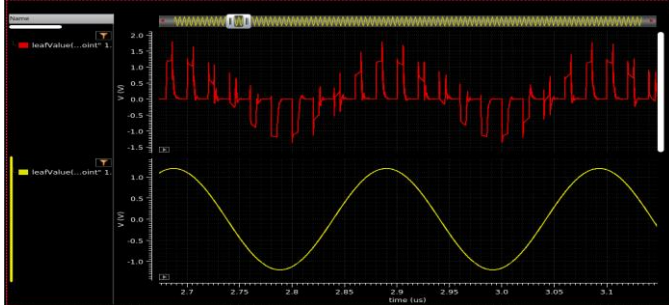
### B. Split-CDAC Array (Analoglib Capacitors + Switches)



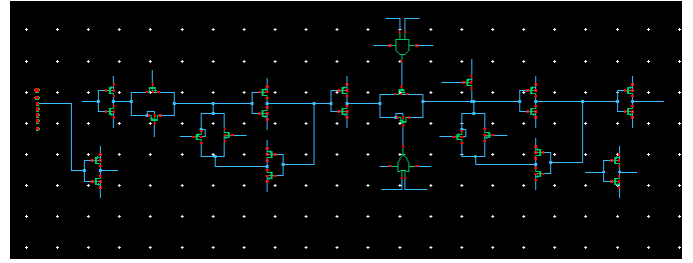
### C. Gate-Bootstrapped Switch (Transistor Level)



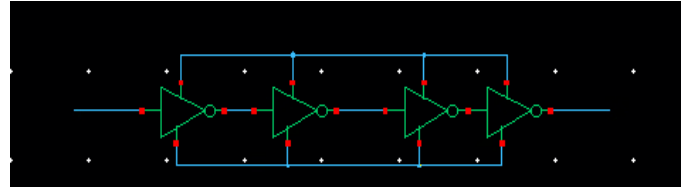
Main Switch size is 150u/0.1u



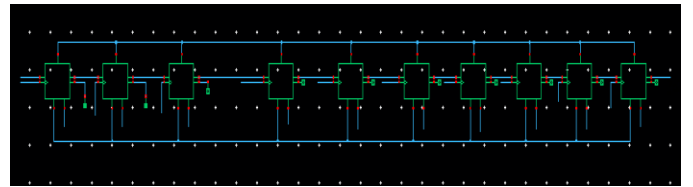
### D. D-FlipFlops (Tranistor-level)



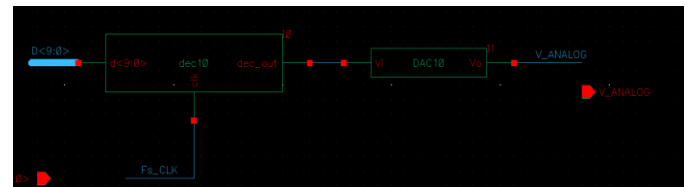
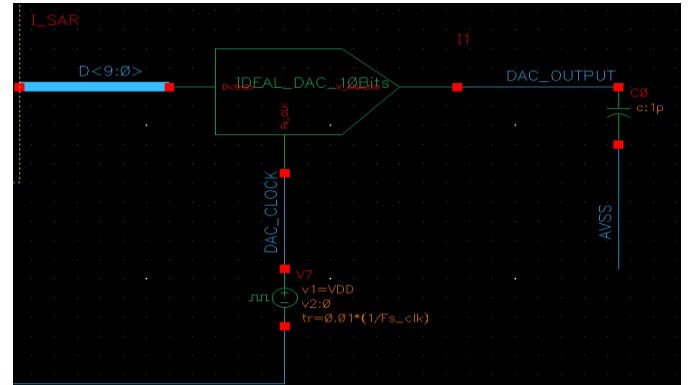
### E. Inverter Chains : 1x → 2x → 4x → 8x (Transistor level)



### F. Ring Counter (Transistor-level)



### G. IdealDAC used for characterization:



Dec10 block converts the D<9:0> into its decimal value, which is then used by the subsequent DAC10 block to convert to its equivalent hold value.

## III. TOP-LEVEL TESTBENCH & SCHEMATIC

### A. Schematic:

Top-Level schematic is shown on Page4. Showing all the blocks inside SAR-ADC (S/H is not part of the schematic).

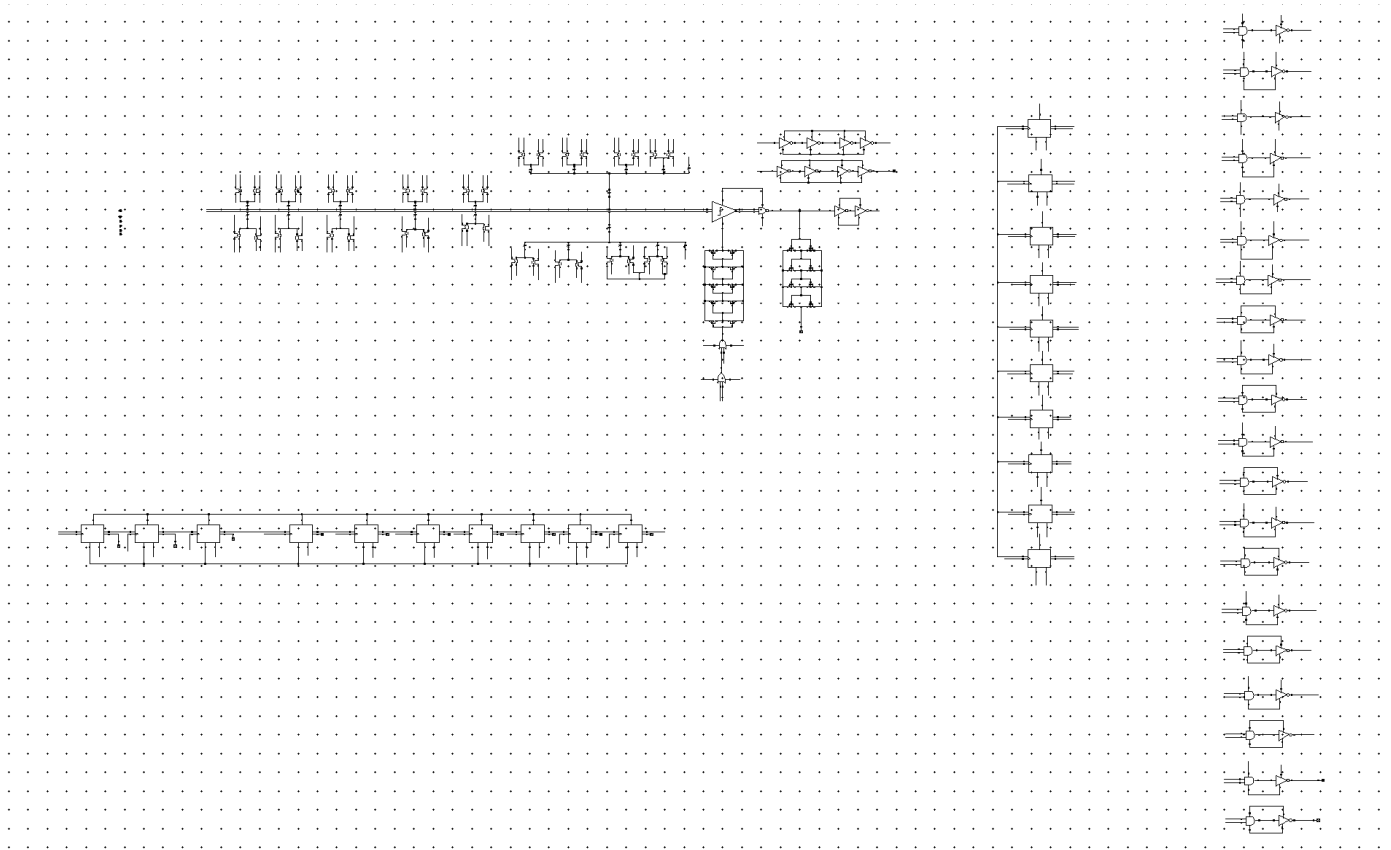


Fig6 Top-Level SAR ADC

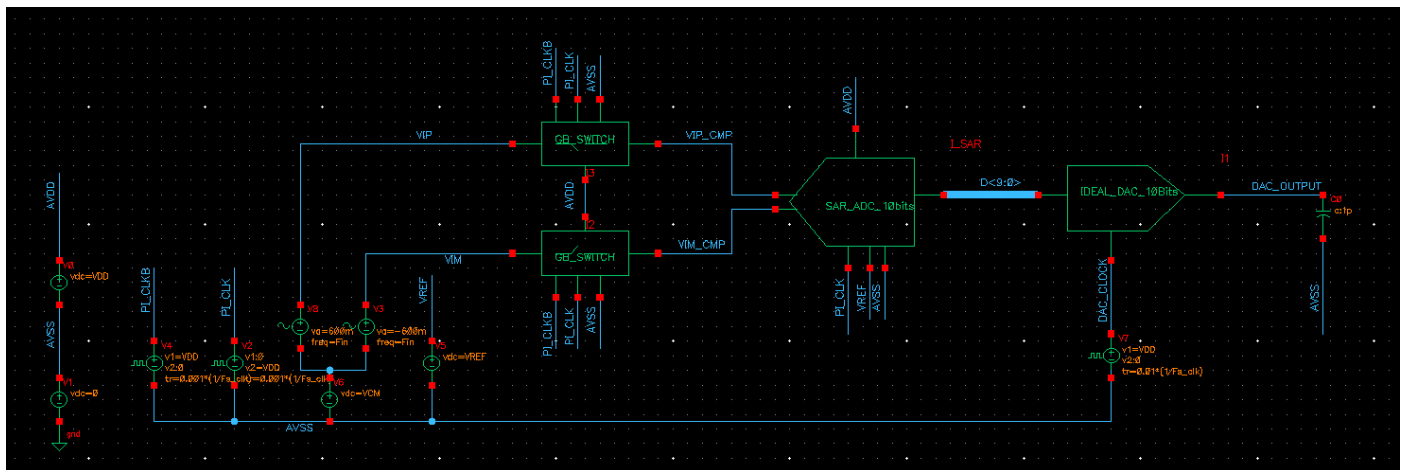


Fig7 Top-Level Testbench (Sine-Wave Testing)

## B. Top-Level Testbench:

Top-Level testbench is shown above. (Fig7)

Sinusoidal Stimulus + S/H switch + SAR-ADC + Ideal-DAC

## C. Sine-Wave Testing for ENOB:

Two Test were performed to characterized ENOB & SNR of the ADC.

$F_s = 50\text{MHz}$

$F_{in} = (m/n) * F_s$ , where  $m$  &  $n$  need to be co-prime.

$n$  represents no. of FFT points which was chosen as 1024.

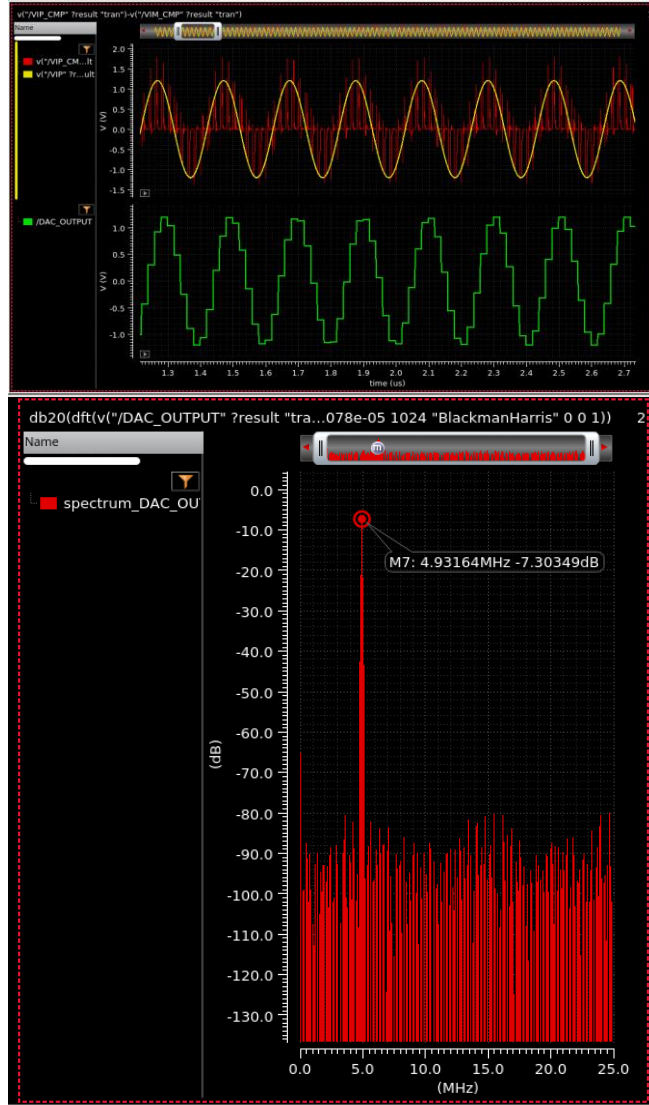
$m=101 \rightarrow F_{in}=4.93\text{ MHz}$

$m=407 \rightarrow F_{in}=19.87\text{ MHz}$

#### IV. SIMULATION RESULTS

- A.  $F_{in}=4.93\text{MHz}$ ,  $F_s=50\text{MHz}$ ,  $V_{REF}=1.2\text{ Volts}$ , input sinusoid amplitude  $600\text{mV}$ . Differential input range was from  $-1.2\text{ Volts}$  to  $1.2\text{ Volts}$

Transient Sim was for  $25\mu\text{s}$  (1024 points were captured and FFT was performed)



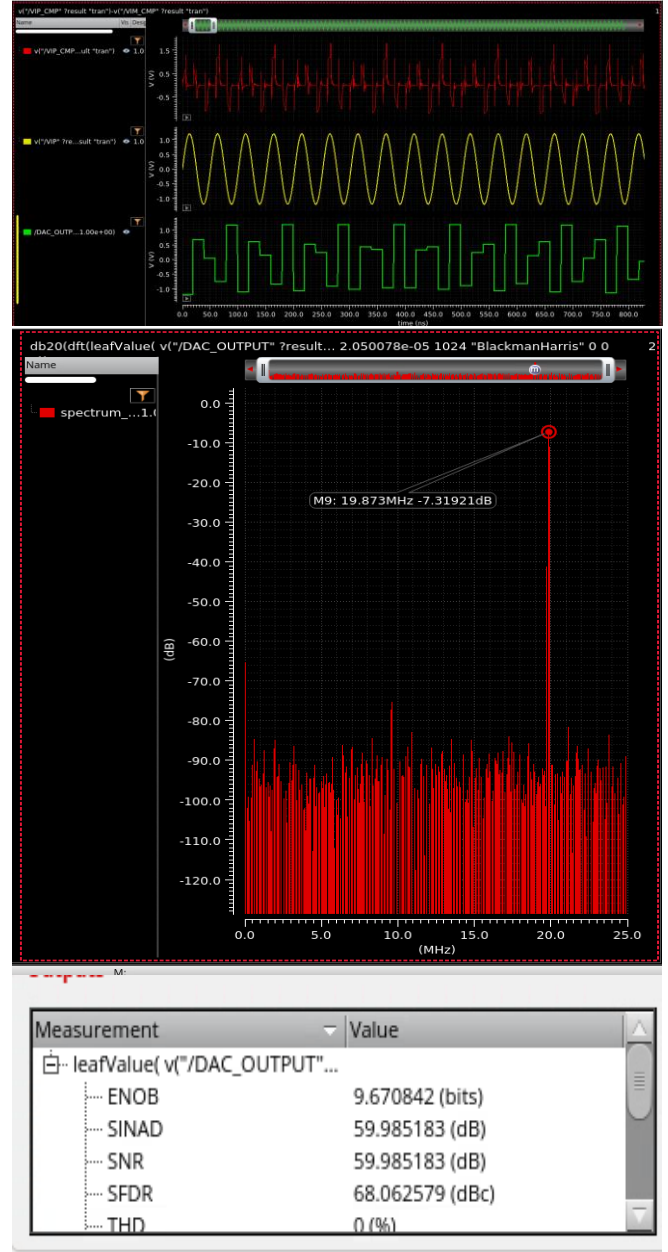
##### Outputs

| Measurement | Value            |
|-------------|------------------|
| /DAC_OUTPUT |                  |
| ENOB        | 9.5511909 (bits) |
| SINAD       | 59.264812 (dB)   |
| SNR         | 59.264812 (dB)   |
| SFDR        | 72.775474 (dBc)  |

ENOB = 9.55 Bits  
 SNR ~ 60dB  
 Power Consumption =  $1.31\text{mW}$

- B.  $F_{in} = 19.87\text{MHz}$ ,  $F_s = 50\text{MHz}$ ,  $V_{REF}=1.2\text{ Volts}$ , input sinusoid amplitude  $600\text{mV}$ . Differential input ranges from  $-1.2$  to  $1.2\text{ Volts}$ .

Transient Sim was run for  $25\mu\text{s}$  (1024 FFT was captured, BlackmanHarris windowing technique was used to reduce the spectral leakage)

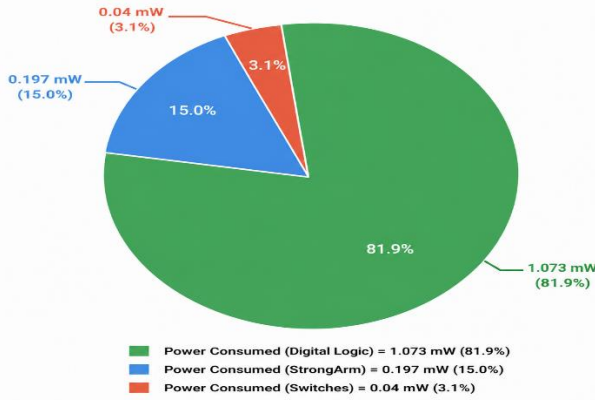


ENOB = 9.67 Bits  
 SNR ~ 60 dB  
 Total Power Consumption (From Supply) =  $1.31\text{ mW}$   
 Power Consumed (StrongArm) =  $0.197\text{ mW}$   
 Power Consumed (Switches) =  $0.04\text{mW}$   
 Power Consumed (SAR Logic) =  $1.073\text{mW}$



## Power Consumption Breakdown

Total Power Consumption = 1.31 mW



## V. CONCLUSIONS

Schreier FOM =  $\text{SNR} + 10 \cdot \log(\text{BW}/\text{Power})$

Schreier FOM = 163 dB

Max Signal BW = 25MHz

Most of the power is consumed in the SAR digital logic due to the dominant  $CV^2f$  switching power. While reducing the sampling frequency lowers this power consumption, it does not improve the overall figure-of-merit (FOM), since the achievable signal bandwidth is also reduced proportionally. Therefore, time-interleaving becomes an effective approach to increase the signal bandwidth without degrading the FOM.

| Specification            | [1]    | [2]  | [3]    | This Work |
|--------------------------|--------|------|--------|-----------|
| Architecture             | SAR    | SAR  | SAR    | SAR       |
| Technology               | 0.13um | 90nm | 90nm   | 90nm      |
| Supply Voltage(V)        | 1.2    | 1    | 1      | 1.2       |
| Sampling Rate(MS/s)      | 50     | 50   | 40     | 50        |
| Resolution               | 10     | 9    | 9      | 10        |
| Sampling Cap(pF)         | 2.5    | 10   | 5      | 32        |
| ENOB                     | 9.18   | 7.8  | 8.56   | 9.55      |
| Power(mW)                | 0.826  | 0.7  | 0.82   | 1.31      |
| Schreier FOM(dB)         | 162    | 154  | 157.16 | 163       |
| Walden FOM(fJ/conv-step) | 29     | 65   | 54     | 33        |

## REFERENCES

- [1] C. -C. Liu, S. -J. Chang, G. -Y. Huang and Y. -Z. Lin, "A 10-bit 50-MS/s SAR ADC With a Monotonic Capacitor Switching Procedure," in *IEEE Journal of Solid-State Circuits*, vol. 45, no. 4, pp. 731-740, April 2010, doi: 10.1109/JSSC.2010.2042254.
- [2] J. Craninckx and G. van der Plas, "A 65fJ/Conversion-Step 0-to-50MS/s 0-to-0.7mW 9b Charge-Sharing SAR ADC in 90nm Digital CMOS," *2007 IEEE International Solid-State Circuits Conference*.
- [3] V. Giannini, P. Nuzzo, V. Chironi, A. Baschiroto, G. Van der Plas and J. Craninckx, "An 820μW 9b 40MS/s Noise-Tolerant Dynamic-SAR ADC in 90nm Digital CMOS," *2008 IEEE International Solid-State Circuits Conference - Digest of Technical Papers*, San Francisco, CA, USA, 2008, pp. 238-610, doi: 10.1109/ISSCC.2008.4523145.
- [4] Y. Zhu et al., "A 10-bit 100-MS/s Reference-Free SAR ADC in 90 nm CMOS," in *IEEE Journal of Solid-State Circuits*, vol. 45, no. 6, pp. 1111-1121, June 2010, doi: 10.1109/JSSC.2010.2048498.
- [5] B. Razavi, "The Bootstrapped Switch [A Circuit for All Seasons]," in *IEEE Solid-State Circuits Magazine*, vol. 7, no. 3, pp. 12-15, Summer 2015, doi: 10.1109/MSSC.2015.2449714.

- [6] B. Razavi, "The StrongARM Latch [A Circuit for All Seasons]," in *IEEE Solid-State Circuits Magazine*, vol. 7, no. 2, pp. 12-17, Spring 2015, doi: 10.1109/MSSC.2015.2418155

## APPENDIX (Verilog A codes):

```

analog begin
    // Keep node from floating before first clock edge
    V(dec_out) <+ 0;

    @(cross(V(clk) - vtrans, +1)) begin
        // sample each bit at the clock edge
        b0 = (V(d[0]) > vtrans) ? 1.0 : 0.0; // LSB
        b1 = (V(d[1]) > vtrans) ? 1.0 : 0.0;
        b2 = (V(d[2]) > vtrans) ? 1.0 : 0.0;
        b3 = (V(d[3]) > vtrans) ? 1.0 : 0.0;
        b4 = (V(d[4]) > vtrans) ? 1.0 : 0.0;
        b5 = (V(d[5]) > vtrans) ? 1.0 : 0.0;
        b6 = (V(d[6]) > vtrans) ? 1.0 : 0.0;
        b7 = (V(d[7]) > vtrans) ? 1.0 : 0.0;
        b8 = (V(d[8]) > vtrans) ? 1.0 : 0.0;
        b9 = (V(d[9]) > vtrans) ? 1.0 : 0.0; // MSB

        // binary decimal
        // 2^0, 2^1, 2^2, ..., 2^9 = 512
        code_val = b0*1.0
                  + b1*2.0
                  + b2*4.0
                  + b3*8.0
                  + b4*16.0
                  + b5*32.0
                  + b6*64.0
                  + b7*128.0
                  + b8*256.0
                  + b9*512.0;

    end

    // drive dec_out with the last latched code (0..1023 V)
    V(dec_out) <+ transition(code_val, tdel, trise, tfall);
end
endmodule

```

// VerilogA for SAR\_ADC, DAC\_output, veriloga

```

`include "constants.vams"
`include "disciplines.vams"

module DAC_output(Vi,Vo);
    electrical Vi,Vo;
    input Vi;
    output Vo;

    parameter tdel=0 from [0:inf];
    parameter trise=1p from [0:inf];
    parameter tfall=1p from [0:inf];

    `define N 10
    real i = (1<<N) ; // i=4;
    real Vr=1.2;
    real dac_output;
    analog begin
        dac_output = (V(Vi)-i)*(1.2/(i*1.0));
        V(Vo) <+ transition(dac_output,tdel,trise,tfall);
    end
endmodule

```